



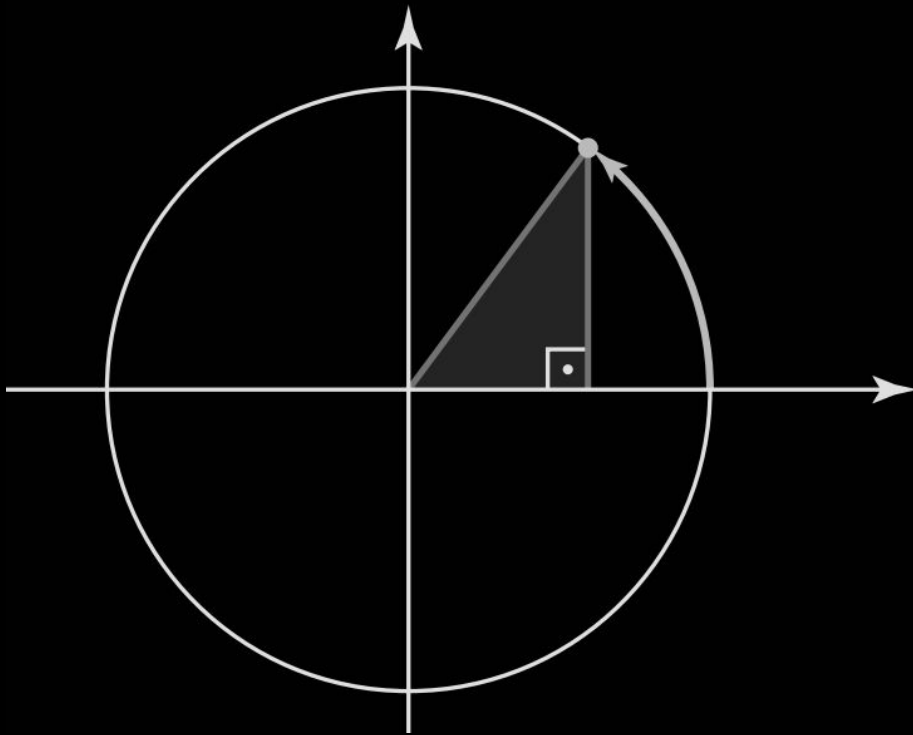
3ª Série

Matemática

Aula 07 - 3º Bim

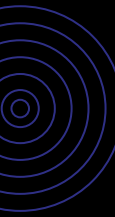
Prof. Diogo Pelaes

Relação fundamental da Trigonometria

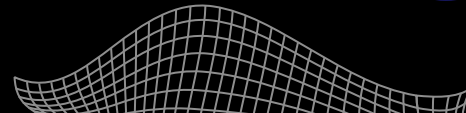
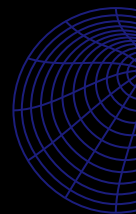
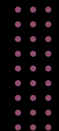




Exercício Resolvido



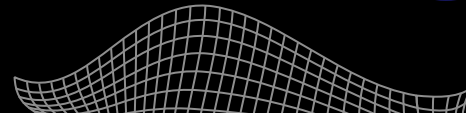
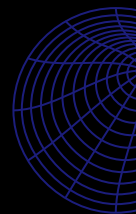
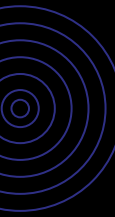
Dado que $\sin \alpha = \frac{1}{3}$, com $\frac{\pi}{2} < \alpha < \pi$, calcule o valor de $\cos \alpha$.





Exercício Resolvido

Determinar os valores de $\sin x$ e de $\cos x$ sabendo que $\sin x = 3 \cos x$ e que $\pi < x < \frac{3\pi}{2}$.

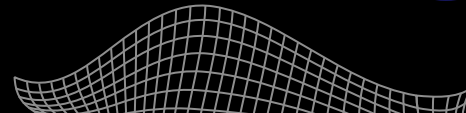
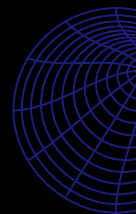


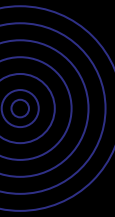


Exercício Resolvido

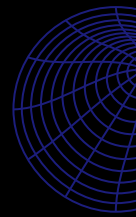
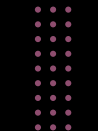
Resolver, em $\mathbb{R}$, a equação na variável x :

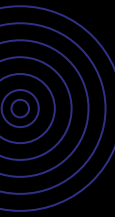
$$x^2 - 2x + \operatorname{sen}^2 \alpha = 0$$



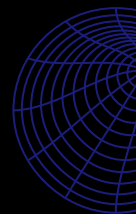
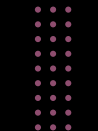


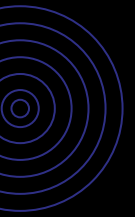
33. Dado que $\operatorname{sen} \alpha = \frac{4}{5}$ e $\frac{\pi}{2} < \alpha < \pi$, calcule o valor de $\cos \alpha$.



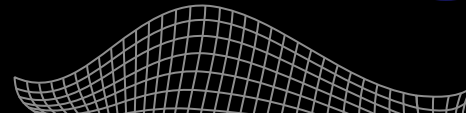
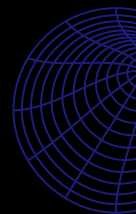


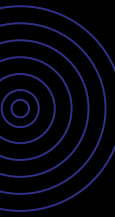
34. Sendo $\cos x = -\frac{1}{3}$ e $\pi < x < \frac{3\pi}{2}$, calcule o valor de $\sin x$.



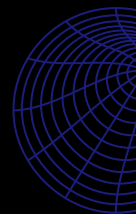
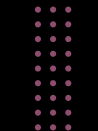


35. Determine $\sin \beta$ e $\cos \beta$ sabendo que $\sin \beta = 2 \cos \beta$
e $\pi < \beta < \frac{3\pi}{2}$.

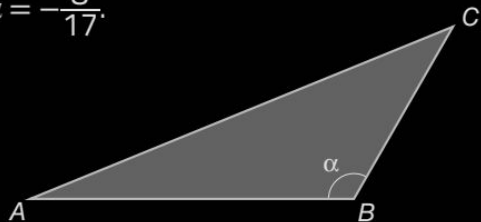




36. Obtenha m , com $m \in \mathbb{R}$, tal que: $\sin x = \frac{m}{4}$ e

$$\cos x = \frac{\sqrt{m+1}}{2}$$


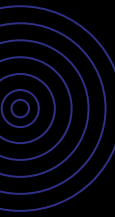
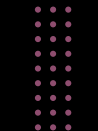
37. No triângulo ABC , a medida do lado $\overline{BC}$ é 34 cm, e o ângulo obtuso $A\hat{B}C$ tem medida α , com $\cos \alpha = -\frac{8}{17}$.

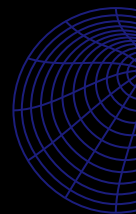


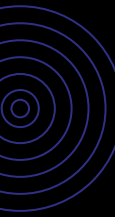
- Copie esse triângulo em seu caderno e desenhe a altura relativa ao lado $\overline{AB}$.
- Calcule a medida h da altura que você desenhou no item a.



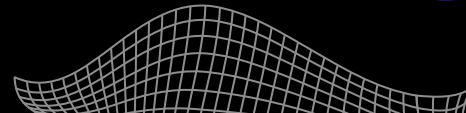
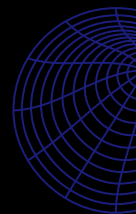
38. Resolva, em $\mathbb{R}$, a equação na variável x :


$$x^2 + 4x + 4 \cos^2 \alpha = 0$$




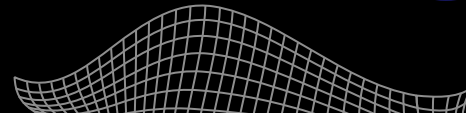
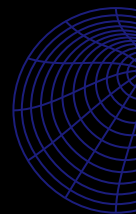
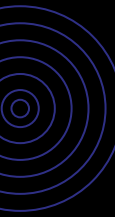


39. Determine o valor do $\cos x$, sabendo que
 $3 \operatorname{sen}^2 x - 4 \operatorname{sen} x + 1 = 0$ e que $0 < x < \frac{\pi}{2}$.

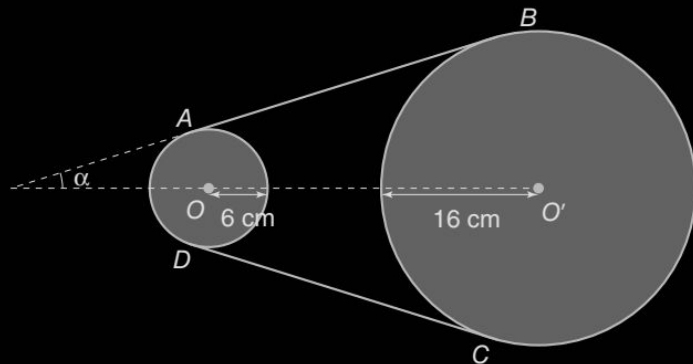




40. Sabendo que $\frac{\pi}{2} < x < \pi$ e que $4 \cos^2 x + 5 \operatorname{sen} x - 5 = 0$,
calcule o valor de $\operatorname{sen} x$.



41. Duas polias de centros O e O' e raios 6 cm e 16 cm, respectivamente, são conectadas por uma correia que as tangencia nos pontos A , B , C e D , conforme mostra a figura a seguir, em que α é a medida do ângulo agudo formado pelas retas $\overleftrightarrow{AB}$ e $\overleftrightarrow{DO}$, com $\cos \alpha = \frac{12}{13}$. Calculam a distância entre os centros das polias.



GABARITO

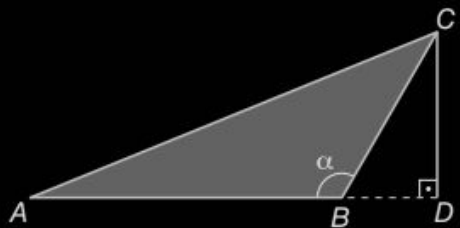
33. $-\frac{3}{5}$

34. $-\frac{2\sqrt{2}}{3}$

35. $\operatorname{sen} \beta = -\frac{2\sqrt{5}}{5}; \cos \beta = -\frac{\sqrt{5}}{5}$

36. $m = 2$

37. a.



b. 30 cm

38. $S = \{-2 + \operatorname{sen} \alpha, -2 - \operatorname{sen} \alpha\}$

39. $\frac{2\sqrt{2}}{3}$

40. $\frac{1}{4}$